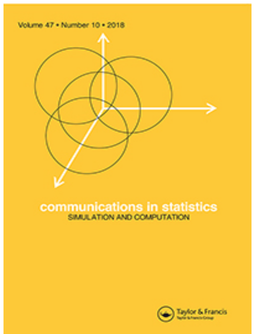

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An extended Maxwell semiparametric regression for censored and uncensored data

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ABSTRACT
We propose a semiparametric regression defined from the generalized odd log-logistic Maxwell distribution under the cubic spline with linear and nonlinear effects for censored and uncensored data. The parameter estimates for this regression are determined using penalized likelihood. Global influence diagnostics and quantile residuals are addressed in new standards. Several Monte Carlo simulations investigate the consistency of the estimates. The usefulness of the new regression is illustrated by means of two applications to real data.

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Censored data; Generalized odd log-logistic family; Maxwell distribution; Penalized likelihood; Semiparametric regression

1. Introduction

Regression is one of the most widely used techniques to model a response variable in terms of explanatory variables. The simplest form of these relationships is linear, and the normal linear model continues to be the most used in applications.

However, the linearity assumption between the response variable and explanatory variables may not be appropriate. It is then interesting to put together a semiparametric regression to evaluate linear and non-linear effects together. For some types of censored data in experiments, it is necessary to take into account these censored observations in the analysis. Recently, several regressions for censored data considering linear and non-linear effects have been investigated. For example, Vanegas and Paula (2015) studied a semiparametric method for joint modeling of the median and skewness, Dhekale et al. (2017) presented application of regressions to tea productivity, Ramires et al. (2018) introduced the semiparametric cure rate survival model based on the sinh Cauchy distribution, Lee and Sison-Mangus (2018) developed a Bayesian semiparametric regression for joint analysis of microbiome data, Vasconcelos et al. (2020) defined two types of extended log-normal regressions and Prataviera, Ortega, Cordeiro, and Cancho (2020) proposed an exponentiated power exponential semiparametric regression.

An example is the relation between *body mass index* (BMI) and patient age when measured before liver transplantation. The BMI is a standard international measure to assess whether or not a person has an ideal weight. It is adopted as a metric by the World Health Organization (WHO) to measure obesity. Overweight and obesity, as indicated by the BMI, are risk factors for diseases like arterial hypertension, coronary artery disease, metabolic syndrome, respiratory diseases, digestive tract diseases, psychiatric disturbances, cancer, osteoarthritis and diabetes mellitus, besides other pathologies considered to be serious public health problems.

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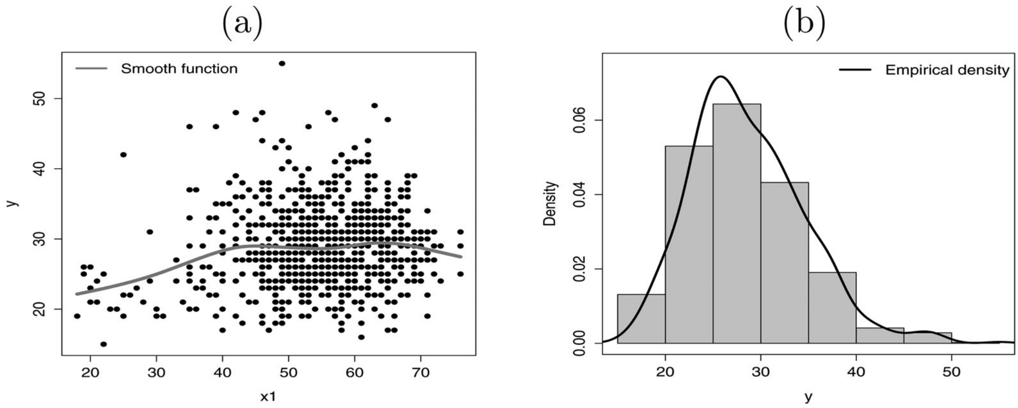

Figure 1. (a) Nonlinear effect of BMI versus age. (b) Histogram for BMI and empirical density.

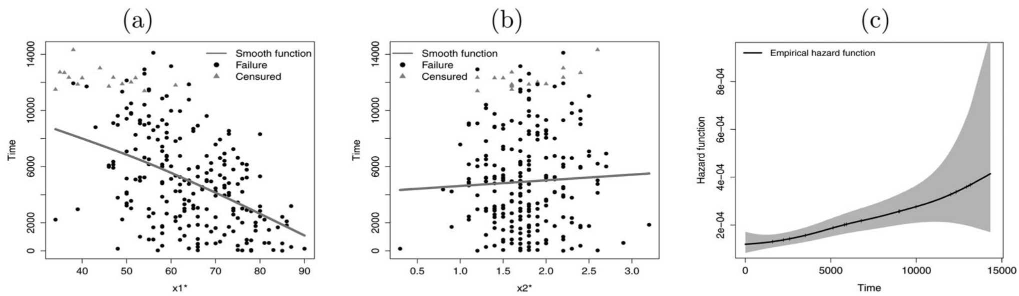

Figure 2. (a) Nonlinear effect of time (in days) versus age. (b) Nonlinear effect of time versus monoclonal protein size. (c) Empirical hazard function for the time.

We analyzed the data set presented in Kelly et al. (2012). Figure 1a shows a nonlinear relationship between the $y=$ BMI (response variable) and $x_1=$ age instead of having of a point cloud segmentation. In turn, Figure 1b shows that the response variable has a certain degree of asymmetry.

A second data set refers to the monoclonal gammopathy of undetermined significance (MGUS) data included in the survival library of the R software. In Figures 2a and Figure 2(b), y denotes the time in days from diagnosis to last follow-up and the covariates: x_1, the age (in years) at diagnosis of MGUS, and x_2, the size of the monoclonal protein spike at diagnosis. These plots indicate that the mentioned problems in the first data set are also presented in the second data set which has censored observations. Thus, the behavior of the empirical hazard function is displayed in Figure 2c.

Recently, Prataviera, Ortega, and Cordeiro (2020) introduced the *generalized odd log-logistic Maxwell* (GOLLMax) distribution with two additional parameters. In this paper, the GOLLMax semiparametric regression is proposed to solve the problems previously described. The rest of the paper is addressed as follows. The GOLLMax parametric regression (for censored and uncensored observations) is introduced in Sec. 2. The GOLLMax semiparametric regression is addressed in Sec. 3 using cubic splines to estimate the nonlinear effects of the covariates in addition to exploratory and diagnostic analysis to assess the suitability of the model. Some simulation results reported in Sec. 4 examine the asymptotic behavior of the maximum likelihood estimates (MLEs) in the new regressions. In Sec. 5, the flexibility of these regressions is illustrated by means of two applications. Finally, some conclusions are offered in Sec. 6.

2. The GOLLMax parametric regression

The probability density function (pdf) of the Maxwell (Max) random variable Z has the form

$$g(z;\mu)=\frac{4}{\sqrt{\pi}}\ \frac{z^2}{\mu^3}\quad \exp\left(-\frac{z^2}{\mu^2}\right),\quad z>0, \tag{1}$$

where z denotes the molecule speed, $\mu>0$ is a scale parameter depending on three quantities (Boltzmann constant, temperature and mass of a molecule) and $\gamma_1(p,u)=\gamma(p,u)/\Gamma(p)$ is the incomplete gamma function ratio.

The cumulative distribution function (cdf) of Z is

$$G(z;\mu)=\gamma_1\left(\frac{3}{2},\frac{z^2}{\mu^2}\right). \tag{2}$$

The nth ordinary moment (for $n>-3$) of Z is

$$E(Z^n)=\frac{2\mu^n}{\sqrt{\pi}}\ \Gamma\left(\frac{n+3}{2}\right),$$

and then $E(Z)=2\mu/\sqrt{\pi}$ and $Var(Z)=(3\pi-8)\mu^2/(2\pi)$. So, the parameter $\mu>0$ is the mean of Z except for a multiplier.

The cdf and pdf of a random variable Y having the GOLLMax distribution are (for $y>0$)

$$F(y;\mu,\sigma,\nu)=\frac{\gamma_1^{\nu\sigma}(3/2,y^2/\mu^2)}{\gamma_1^{\nu\sigma}(3/2,y^2/\mu^2)+\left[1-\gamma_1^{\sigma}(3/2,y^2/\mu^2)\right]^{\nu}} \tag{3}$$

and

$$f(y;\mu,\sigma,\nu)=\frac{4\ \nu\ \sigma\ y^2}{\sqrt{\pi}\ \mu^3}\quad \exp\left(-\frac{y^2}{\mu^2}\right)\frac{\gamma_1^{\nu\sigma-1}(3/2,y^2/\mu^2)\left[1-\gamma_1^{\sigma}(3/2,y^2/\mu^2)\right]^{\nu-1}}{\left\{\gamma_1^{\nu\sigma}(3/2,y^2/\mu^2)+\left[1-\gamma_1^{\sigma}(3/2,y^2/\mu^2)\right]^{\nu}\right\}^2}, \tag{4}$$

respectively, where $\nu>0$ and $\sigma>0$ are two extra shape parameters, and $\mu>0$ works as a scale parameter. Henceforth, $Y\sim$GOLLMax(ν,σ,μ) denotes a random variable with cdf (3).

The following well-known distributions are special cases of Equation (4). The exponentiated Max (EMax) and odd log-logistic Max (OLLMax) distributions correspond to $\nu=1$ and $\sigma=1$, respectively. The Max distribution is a basic exemplar when $\sigma=\nu=1$.

The quantile function (qf) of Y (for $0\le u\le 1$) follows by inverting (3)

$$y=Q_{\text{Max}}(w;\mu)=\mu\ \sqrt{\gamma_1^{-1}(3/2,w)}, \tag{5}$$

where $Q_{\text{Max}}(w;\mu)=G^{-1}(w;\mu)$ is the Max qf,

$$w=\left[\frac{u^{\frac{1}{\nu}}}{(1-u)^{\frac{1}{\nu}}+u^{\frac{1}{\nu}}}\right]^{\frac{1}{\sigma}},$$

and $\gamma_1^{-1}(3/2,w)$ is the inverse of the upper gamma regularized function. For more details, see http://functions.wolfram.com/GammaBetaErf/InverseGammaRegularized/.

Let $Y_1,\ldots,Y_n$ be independent random variables, each with density function (4) but varying the scale parameter μ_i, and $\mathbf{x}_i^{*T}=(x_{i1}^*,\ldots,x_{ip}^*)$ be a vector of fixed explanatory variables for the ith observation. Consider the systematic component

$$g(\mu_i)=\log(\mu_i)=\mathbf{x}_i^{*T}\boldsymbol{\beta}, \tag{6}$$

where $\boldsymbol{\beta}=(\beta_{11},\ldots,\beta_{1p})^T$ is the parameter vector of dimension p.

The GOLLMax parametric regression is defined by (4) and (6). The total log-likelihood function for the vector $\boldsymbol{\psi} = (\boldsymbol{\beta}^T, \sigma, \nu)^T$ from an observed sample $(y_1, \mathbf{x}_1^*), \ldots, (y_n, \mathbf{x}_n^*)$ can be expressed as

$$\begin{aligned} l(\boldsymbol{\psi}) = {} & n \log\left(\frac{4\ \sigma\ \nu}{\sqrt{\pi}}\right) + \sum_{i=1}^{n} \log\left(\frac{y_i^2}{\mu_i^3}\right) - \sum_{i=1}^{n} \left(\frac{y_i^2}{\mu_i^2}\right) \\ & + (\sigma\ \nu - 1) \sum_{i=1}^{n} \log\left[\gamma_1(3/2, y_i^2/\mu_i^2)\right] + (\nu - 1) \sum_{i=1}^{n} \log\left[1 - \gamma_1(3/2, y_i^2/\mu_i^2)\right] \\ & - 2 \sum_{i=1}^{n} \log\left\{\gamma_1^{\sigma\nu}(3/2, y_i^2/\mu_i^2) + \left[1 - \gamma_1^{\sigma}(3/2, y_i^2/\mu_i^2)\right]^{\nu}\right\}. \end{aligned} \tag{7}$$

The proof of the usual maximum likelihood regularity conditions analytically even for special parametric models without censoring observations is extremely difficult because of the incomplete gamma function. There are sufficient conditions for the existence of the MLEs such as compactness of the parameter space and the concavity of the log-likelihood function given by (7), but they can exist even when these conditions are not satisfied. In general, there is no explicit solution for the estimates from maximizing (7), but we can study numerically conditions on their existence and uniqueness for very special models (when there are no censoring observations) by examining the intervals of variation of the score functions. The large sample properties of the estimators follow from the regularity conditions. However, this investigation is not the main objective of this paper.

For censored data, let Y_i be the lifetime and C_i be the censoring time for the ith individual. The observed time is $y_i = \min\{Y_i, C_i\}$ and its contribution to the log-likelihood function is

$$\begin{aligned} l(\boldsymbol{\psi}) = {} & r \log\left(\frac{4\ \sigma\ \nu}{\sqrt{\pi}}\right) + \sum_{i \in F} \log\left(\frac{y_i^2}{\mu_i^3}\right) - \sum_{i \in F} \left(\frac{y_i}{\mu_i}\right)^2 \\ & + (\sigma\ \nu - 1) \sum_{i \in F} \log\left[\gamma_1(3/2, y_i^2/\mu_i^2\right] + (\nu - 1) \sum_{i \in F} \log\left[1 - \gamma_1^{\sigma}(3/2, y_i^2/\mu_i^2)\right] \\ & - 2 \sum_{i \in F} \log\left\{\gamma_1^{\sigma\nu}(3/2, y_i^2/\mu_i^2) + \left[1 - \gamma_1^{\sigma}(3/2, y_i^2/\mu_i^2)\right]^{\nu}\right\} \\ & + \sum_{i \in C} \log\left\{1 - \frac{\gamma_1^{\sigma\nu}(3/2, y_i^2/\mu_i^2)}{\gamma_1^{\sigma\nu}(3/2, y_i^2/\mu_i^2) + [1 - \gamma_1^{\sigma}(3/2, y_i^2/\mu_i^2)]^{\nu}}\right\}, \end{aligned} \tag{8}$$

where r is the number of failures (uncensored observations), and F and C are the sets of individuals with lifetime and censoring time, respectively.

3. The GOLLMax semiparametric regression

In some cases, covariates have non-linear effects with the response variable. So, it is necessary to extend the systematic component (6) to

$$g(\mu_i) = \mathbf{x}_i^{*T} \boldsymbol{\beta} + h_1(x_{i1}) + \cdots + h_J(x_{iJ}), \tag{9}$$

where $h_j(\cdot)$'s are smooth functions of the covariates $\mathbf{x}_{ij}$ (for $j = 1, \ldots, J, i = 1, \ldots, n$), and assume that $g(\cdot)$ is a known one-to-one continuously twice differentiable function.

In this paper, the approximation of $h_j(\cdot)$ is by cubic spline using the gamlss package of the R software. Such smoothing functions are expressed as random effects, i.e., $h_j(\cdot) = Z_j \boldsymbol{\varphi}_j$, where Z_j is the $(n \times q_j)$ known baseline matrix and $\boldsymbol{\varphi}_j$ is the q_j-dimensional vector of unknown parameters.

The fixed and random effects $\boldsymbol{\psi} = (\boldsymbol{\beta}^T, \sigma, \nu)^T$ and $\boldsymbol{\varphi} = (\boldsymbol{\varphi}_1, ..., \boldsymbol{\varphi}_J)^T$ in model (9), respectively, can be calculated by maximizing the penalized log-likelihood function

$$l_p(\boldsymbol{\omega}) = l(\boldsymbol{\psi}) - \frac{1}{2}\sum_{j=1}^{J} \lambda_j \boldsymbol{\varphi}_j^T\ \mathbf{P}_j \boldsymbol{\varphi}_j, \tag{10}$$

for $\boldsymbol{\omega} = (\boldsymbol{\beta}^T, \sigma, \nu, \boldsymbol{\varphi}^T, \boldsymbol{\lambda}^T)^T$, where $l(\boldsymbol{\psi})$ can be (7) or (8), λ_j is the unknown smoothing parameter and P_j is a symmetric matrix that may depend on a vector of smoothing parameters and chosen number of knots. The maximization of (10) corresponds to the smoothing cubic spline with equidistant knots for distinct x-variable values; for more details, see Green and Silverman (1993), Ruppert, Wand, and Carroll (2003) and Wood (2017). Another measure of interest is the effective degrees of freedom ($df^{\dagger}$) corresponding to the non-parametric component. The smoothing parameters can be fixed or estimated from the data. Some methods are proposed in the literature, for example, by the generalized cross-validation method (Wood, 2017). However, in this work, due to the direct relationship between $df^{\dagger}$ and λ_j to penalize overfitting in proportion to the complexity of the model, the Akaike information criterion (AIC) is adopted.

The maximization of (10) can be performed in the `RS` algorithm (Stasinopoulos and Rigby 2007; Stasinopoulos et al. 2017). The function `cs()` is used to assign the arguments to make the adjustment via `gamlss`. Thus, in Sec. 5, the `cs(·)` function in regression structures is denoted by $cs(x_{ij}, df^{\dagger})$, where x_{ij} is the jth covariate for the additive term and $df^{\dagger}$ is the degrees of freedom. The effective degree of freedom for the systematic component in μ for an explanatory variable x is $df_{\mu} = df^{\dagger} + 2$, where the other two additional degrees of freedom are related to the linear terms (Voudouris et al. 2012). Finally, the total degrees of freedom of the adjusted model, say df, collectively considers the additive terms represented by the functions $h_j(\cdot)$ and the parametric terms, i.e., $df = df_{\mu} + df_{\sigma} + df_{\nu}$ denotes the degrees of freedom used to model μ, σ and ν, respectively.

3.1. Model selection

The global deviance (GD), $GD = -2l_p(\hat{\boldsymbol{\omega}})$, and the generalized Akaike information criterion (GAIC), $GAIC(k) = GD + k \times df$, where $l_p(\hat{\boldsymbol{\omega}})$ is the penalized log-likelihood function, df is the total degrees of freedom of the fitted regression and k is the penalty for each degree of freedom, are used for selecting the appropriate regression. For $k=2$ and $k = \log(n)$, it follows the Akaike information criterion (AIC) and Bayesian information criterion (BIC), respectively.

3.2. Influence and residual analysis

Let $\hat{\boldsymbol{\omega}}_{(i)}$ be the MLE of $\boldsymbol{\omega}$ for the sample excluding the ith observation. The difference between $\hat{\boldsymbol{\omega}}$ and $\hat{\boldsymbol{\omega}}_{(i)}$ on the log-likelihood scale (Cook and Weisberg 1982)

$$LD_i = 2\left[l_p(\hat{\boldsymbol{\omega}}) - l_p(\hat{\boldsymbol{\omega}}_{(i)})\right], \tag{11}$$

evaluates the influence of the ith observation on the estimates.

The quantile residuals (qrs) (Dunn and Smyth 1996) are used to verify possible deviations from the model assumptions. For the GOLLMax semiparametric regression, they are

$$\hat{rq}_i = \Phi^{-1}\left\{\frac{\gamma_1^{\hat{\nu}\hat{\sigma}}(3/2, y^2/\hat{\mu}_i^2)}{\gamma_1^{\hat{\nu}\hat{\sigma}}(3/2, y^2/\hat{\mu}_i^2) + \left[1 - \gamma_1^{\hat{\sigma}}(3/2, y^2/\hat{\mu}_i^2)\right]\hat{\nu}}\right\}, \tag{12}$$

where $\Phi^{-1}(\cdot)$ is the standard normal qf.

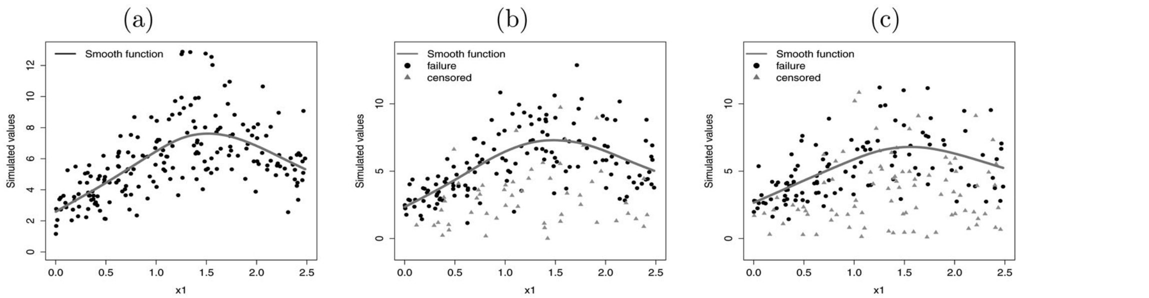

Figure 3. Plots of the simulated values for $n = 200$. (a) Nonlinear effect for scenario 1a. (b) Nonlinear effect for scenario 1b. (c) Nonlinear effect for scenario 1c.

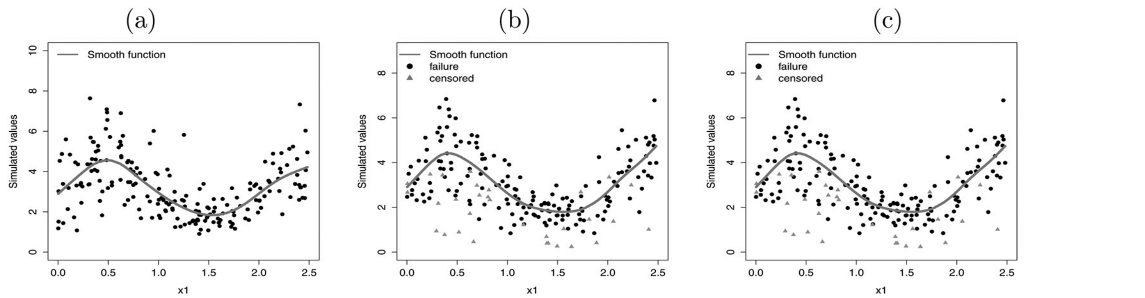

Figure 4. Plots of the simulated values for $n = 200$. (a) Nonlinear effect for scenario 2a. (b) Nonlinear effect for scenario 2b. (c) Nonlinear effect for scenario 2c.

4. Simulation study for the semiparametric regression

The performance of the new semiparametric regression is examined by Monte Carlo simulations for censored and uncensored data with three sample sizes ($n = 80$, 200, 450) using the `R` software with `gamlss` packages by the `RS` method.

Two scenarios with systematic components are defined from (9):

- For the first, $\mu_i = \exp\left\{\beta_{11}x_{i1}^* + \beta_{12}x_{i2}^* + h_1(x_{i1})\right\}$, where $h_1(x_{i1}) = \sin(x_{i1})$. The functional shape of h_1 is displayed in Figures 3a, b and c for scenarios 1a, 1b and 1c, respectively.
- For the second, $\mu_i = \exp\left\{\beta_{11}x_{i1}^* + \beta_{12}x_{i2}^* + h_1(x_{i1})\right\}$, where $h_1(x_{i1}) = 0.45[\sin(\pi\ x_{i1})]$. The functional shape of h_1 is reported in Figures 4a, b and c for scenarios 2a, 2b and 2c, respectively.

For both scenarios $\sigma = \exp(\beta_{20}), \nu = \exp(\beta_{30})$, considering (a) without censored values, (b) with a percentage of censored values approximately 25% and (c) with a percentage of censored values approximately 45%. The associated coefficients for the scenarios are: $\beta_{11} = 0.20, \beta_{12} = -0.35, \beta_{20} = 0.25$ and $\beta_{30} = 0.40$. The explanatory variables are $x_{i1}{}^* \sim$Normal (5,0.5), $x_{i2}{}^* \sim$Uniform (0,1) and $x_{i1} \sim$Uniform (0,2.5), respectively, for $i = 1, \ldots, n$. For all scenarios, $df^\dagger = 3$ which is the default value of the `cs(·)` function.

The GOLLMax semiparametric regression is simulated under the algorithm:

1. Generate the variables x_{i1}^*, x_{i2}^* and x_{i1}.
2. For uncensored values of the scenarios:
   - Generate the values $y_i \sim$GOLLMax(μ_i, σ, ν) from the systematic component (6).
3. For censored values of the scenarios:
   - Generate the values $y_i^* \sim$GOLLMax(μ_i, σ, ν) from the systematic component (6).

Table 1. Results from the (parametric and semiparametric) regressions for scenario 1a.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
80	β_{11}	0.199	−0.001	0.004	β_{11}	0.115	−0.085	0.012
	β_{12}	−0.352	−0.002	0.012	β_{12}	−0.388	−0.038	0.015
	β_{20}	0.239	−0.011	0.007	β_{20}	0.300	0.050	0.113
	β_{30}	0.476	0.076	0.014	β_{30}	0.134	−0.266	0.113
200	β_{11}	0.198	−0.002	0.001	β_{11}	0.214	0.014	0.002
	β_{12}	−0.348	0.002	0.004	β_{12}	−0.317	0.033	0.006
	β_{20}	0.221	−0.029	0.005	β_{20}	0.250	0.000	0.040
	β_{30}	0.440	0.040	0.006	β_{30}	0.171	−0.229	0.068
450	β_{11}	0.201	0.001	0.001	β_{11}	0.207	0.007	0.001
	β_{12}	−0.350	0.000	0.002	β_{12}	−0.359	−0.009	0.002
	β_{20}	0.216	−0.034	0.003	β_{20}	0.258	0.008	0.018
	β_{30}	0.430	0.030	0.003	β_{30}	0.180	−0.220	0.055

Table 2. Results from the (parametric and semiparametric) regressions for scenario 1 b.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
80	β_{11}	0.197	−0.003	0.006	β_{11}	0.112	−0.088	0.015
	β_{12}	−0.355	−0.005	0.004	β_{12}	−0.389	−0.039	0.019
	β_{20}	0.232	−0.018	0.003	β_{20}	0.168	−0.082	0.036
	β_{30}	0.488	0.088	0.016	β_{30}	0.222	−0.178	0.049
200	β_{12}	0.199	−0.001	0.002	β_{11}	0.216	0.016	0.002
	β_{12}	−0.345	0.005	0.006	β_{12}	−0.318	0.032	0.008
	β_{20}	0.218	−0.032	0.002	β_{20}	0.155	−0.095	0.019
	β_{30}	0.447	0.047	0.005	β_{30}	0.236	−0.164	0.033
450	β_{11}	0.200	0.000	0.001	β_{11}	0.205	0.005	0.001
	β_{12}	−0.352	0.002	0.002	β_{12}	−0.361	−0.011	0.003
	β_{20}	0.211	−0.039	0.002	β_{20}	0.160	−0.090	0.013
	β_{30}	0.434	0.034	0.002	β_{30}	0.245	−0.155	0.027

Table 3. Results from the (parametric and semiparametric) regressions for scenario 1c.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
80	β_{11}	0.196	−0.004	0.009	β_{11}	0.107	−0.093	0.020
	β_{12}	−0.362	−0.012	0.021	β_{12}	−0.395	−0.045	0.030
	β_{20}	0.234	−0.016	0.003	β_{20}	0.141	−0.109	0.018
	β_{30}	0.496	0.096	0.021	β_{30}	0.267	−0.133	0.028
200	β_{11}	0.198	−0.002	0.003	β_{11}	0.216	0.016	0.003
	β_{12}	−0.344	0.006	0.008	β_{12}	−0.322	0.028	0.010
	β_{20}	0.217	−0.033	0.002	β_{20}	0.136	−0.114	0.015
	β_{30}	0.450	0.050	0.006	β_{30}	0.271	−0.129	0.021
450	β_{11}	0.197	−0.003	0.001	β_{11}	0.194	−0.006	0.002
	β_{12}	−0.355	−0.005	0.003	β_{12}	−0.378	−0.028	0.005
	β_{20}	0.210	−0.040	0.002	β_{20}	0.135	−0.115	0.015
	β_{30}	0.434	0.034	0.003	β_{30}	0.261	−0.139	0.021

- Generate $c_i \sim \text{Uniform}(0, \zeta)$, where ζ denotes the proportion of censored observations.
- Set $y_i = \min(y_i^*, c_i)$.
- Define a vector δ of dimension n which receives one if $(y_i^* \leq c_i)$ and zero otherwise.

For each of the 1, 000 simulations, we calculate the average estimates (AEs), biases and MSEs which are reported in Tables 1–6 for the parametric and semiparametric regressions. Based on these simulation results, we check how much the inclusion of an additive term affects the estimation of the other fixed parameters. For the semiparametric regression, we verify that the MSEs of the MLEs of β_{11}, β_{12}, β_{20} and β_{30} for the current scenarios decay toward zero and the AEs of the

Table 4. Results from the (parametric and semiparametric) regressions for scenario 2a.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
	β_{11}	0.186	−0.014	0.004	β_{11}	0.240	0.040	0.006
	β_{12}	−0.366	−0.016	0.012	β_{12}	−0.301	0.049	0.017
80	β_{20}	0.233	−0.017	0.007	β_{20}	0.553	0.303	0.356
	β_{30}	0.450	0.050	0.010	β_{30}	−0.174	−0.574	0.419
	β_{11}	0.197	−0.003	0.001	β_{11}	0.192	−0.008	0.002
	β_{12}	−0.353	−0.003	0.005	β_{12}	−0.436	−0.086	0.013
200	β_{20}	0.216	−0.034	0.006	β_{20}	0.637	0.387	0.301
	β_{30}	0.416	0.016	0.004	β_{30}	0.294	−0.694	0.531
	β_{11}	0.200	0.000	0.001	β_{11}	0.187	−0.013	0.001
	β_{12}	−0.361	−0.011	0.002	β_{12}	−0.393	−0.043	0.004
450	β_{20}	0.212	−0.038	0.004	β_{20}	0.476	0.226	0.108
	β_{30}	0.406	0.006	0.002	β_{30}	−0.172	−0.572	0.347

Table 5. Results from the (parametric and semiparametric) regressions for scenario 2 b.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
	β_{11}	0.184	−0.016	0.006	β_{11}	0.242	0.042	0.008
	β_{12}	−0.370	−0.020	0.014	β_{12}	−0.316	0.034	0.019
80	β_{20}	0.219	−0.031	0.004	β_{20}	0.377	0.127	0.198
	β_{30}	0.457	0.057	0.010	β_{30}	−0.072	−0.472	0.298
	β_{11}	0.197	−0.003	0.002	β_{11}	0.191	−0.009	0.002
	β_{12}	−0.354	0.004	0.005	β_{12}	−0.437	−0.087	0.015
200	β_{20}	0.205	−0.045	0.003	β_{20}	0.481	0.231	0.183
	β_{30}	0.420	0.020	0.003	β_{30}	−0.194	0.594	0.401
	β_{11}	0.197	−0.003	0.001	β_{11}	0.186	−0.014	0.001
	β_{12}	−0.363	−0.013	0.002	β_{12}	−0.394	−0.044	0.005
450	β_{20}	0.200	−0.050	0.003	β_{20}	0.383	0.133	0.068
	β_{30}	0.408	0.008	0.001	β_{30}	−0.011	−0.510	0.280

Table 6. Results from the (parametric and semiparametric) regressions for scenario 2c.

	Semiparametric				Parametric			
n	Parameter	AE	Bias	MSE	Parameter	AE	Bias	MSE
	β_{11}	0.223	0.023	0.008	β_{11}	0.265	0.065	0.016
	β_{12}	−0.341	0.009	0.020	β_{12}	−0.372	0.022	0.028
80	β_{20}	0.218	−0.032	0.003	β_{20}	0.167	−0.083	0.091
	β_{30}	0.451	0.051	0.010	β_{30}	0.051	−0.349	0.166
	β_{11}	0.192	−0.008	0.003	β_{11}	0.140	−0.060	0.007
	β_{12}	−0.367	−0.017	0.008	β_{12}	−0.443	−0.093	0.020
200	β_{20}	0.203	−0.047	0.003	β_{20}	0.199	−0.051	0.050
	β_{30}	0.412	0.012	0.003	β_{30}	0.046	−0.354	0.150
	β_{11}	0.194	−0.006	0.001	β_{11}	0.215	0.015	0.002
	β_{12}	−0.354	−0.004	0.003	β_{12}	−0.307	0.043	0.007
450	β_{20}	0.199	−0.051	0.002	β_{20}	0.351	0.101	0.052
	β_{30}	0.401	0.001	0.001	β_{30}	−0.076	−0.476	0.246

parameters tend to be closer to the true parameter values when the sample size n increases (for censored and uncensored values). However, for the parametric regression, such measures do not exhibit the same behavior. Also, we verify their average lengths (ALs) and the empirical coverage probabilities (CPs), say C(ψ), corresponding to the 95% confidence intervals calculated from the simulations for the parametric regression. The results are given in Tables 7, 8 and 9 for 0%, 25% and 45% censored, respectively.

The behavior of the nonlinear effects in the simulations (scenarios a, b and c) is reported in the Appendix. Figures 19–24 display the generated and fitted effects for the parametric and

Table 7. AEs, Biases, MSEs, ALs and CPs of the parameters for the GOLLMax parametric regression for 0% censored.

n	Parameter	AEs	Biases	MSEs	ALs	$C(\psi)$
80	β_{10}	1.386	−0.064	0.228	2.127	0.928
	β_{11}	0.200	0.000	0.004	0.239	0.937
	β_{12}	−0.349	0.001	0.011	0.410	0.952
	β_{20}	0.693	0.443	1.251	4.383	0.872
	β_{30}	0.250	−0.150	0.308	2.458	0.862
200	β_{10}	1.450	0.000	0.101	1.490	0.950
	β_{11}	0.198	−0.002	0.001	0.150	0.958
	β_{12}	−0.350	0.000	0.004	0.259	0.952
	β_{20}	0.411	0.161	0.564	3.181	0.891
	β_{30}	0.355	−0.045	0.171	1.841	0.885
450	β_{10}	1.450	0.000	0.060	1.059	0.957
	β_{11}	0.200	0.000	0.001	0.099	0.961
	β_{12}	−0.349	0.001	0.002	0.172	0.946
	β_{20}	0.325	0.075	0.310	2.369	0.911
	β_{30}	−0.384	−0.016	0.103	1.373	0.903

Table 8. AEs, Biases, MSEs, ALs and CPs of the parameters for the GOLLMax parametric regression for 25% censored.

n	Parameter	AEs	Biases	MSEs	ALs	$C(\psi)$
80	β_{10}	1.376	−0.074	0.282	2.430	0.917
	β_{11}	0.199	−0.001	0.006	0.286	0.941
	β_{12}	−0.349	0.001	0.017	0.494	0.934
	β_{20}	0.775	0.525	1.609	4.896	0.867
	β_{30}	0.225	−0.175	0.384	2.753	0.861
200	β_{10}	1.421	−0.029	0.133	1.714	0.946
	β_{11}	0.201	0.001	0.002	0.179	0.950
	β_{12}	−0.354	−0.004	0.006	0.309	0.951
	β_{20}	0.487	0.237	0.667	3.551	0.890
	β_{30}	0.318	−0.082	0.200	2.064	0.885
450	β_{10}	1.470	0.020	0.079	1.286	0.945
	β_{11}	0.198	−0.002	0.001	0.119	0.950
	β_{12}	−0.349	0.001	0.003	0.205	0.955
	β_{20}	0.316	0.066	0.379	2.754	0.910
	β_{30}	0.391	−0.009	0.126	1.628	0.903

Table 9. AEs, biases, MSEs, ALs and CPs of the parameters for the GOLLMax parametric regression for 45% censored.

n	Parameter	AEs	Biases	MSEs	ALs	$C(\psi)$
80	β_{10}	1.377	−0.073	0.349	2.748	0.932
	β_{11}	0.200	0.000	0.008	0.317	0.932
	β_{12}	−0.352	−0.002	0.023	0.548	0.935
	β_{20}	0.804	0.554	1.808	5.426	0.861
	β_{30}	0.215	−0.185	0.436	3.089	0.859
200	β_{10}	1.415	−0.035	0.150	1.841	0.948
	β_{11}	0.201	0.001	0.002	0.197	0.950
	β_{12}	−0.356	−0.006	0.008	0.341	0.948
	β_{20}	0.509	0.259	0.717	3.743	0.885
	β_{30}	0.308	−0.092	0.218	2.187	0.867
450	β_{10}	1.462	0.012	0.086	1.323	0.947
	β_{11}	0.199	−0.001	0.001	0.131	0.943
	β_{12}	−0.352	−0.002	0.003	0.226	0.950
	β_{20}	0.339	0.089	0.393	2.814	0.916
	β_{30}	0.378	−0.022	0.128	1.663	0.913

semiparametric regressions. They also include the box plots of the GD, AIC and BIC statistics from 1,000 simulations for both regressions. The nonlinear effects are very close to the true shape for large n as shown in Figures 3 and 4. If the percentage of censored observations increases, it tends to show slight deviations in the estimated curves, but without major compromises from the current estimates. Further, the semiparametric regression presents the lowest GD, AIC and BIC

Table 10. Estimates of λ and AIC for the fitted GOLLMax semiparametric regression for scenarios 1a, 1 b and 1c with $n = 200$.

	scenario 1a		scenario 1b		scenario 1c	
$df^{\dagger}$	λ	AIC	λ	AIC	λ	AIC
1	0.191	750.27	0.197	591.95	0.197	455.39
2	0.038	740.15	0.038	579.63	0.039	406.42
3	**0.012**	**739.95**	**0.012**	**579.17**	**0.012**	**405.54**
4	0.004	740.47	0.004	579.75	0.004	406.20
5	0.002	741.10	0.002	580.46	0.002	409.21
6	0.001	741.73	0.001	581.18	0.001	407.86
7	0.001	742.37	0.001	581.90	0.001	408.76
8	0.000	742.99	0.000	582.61	0.000	409.64
9	0.000	743.62	0.000	583.32	0.000	410.70
10	0.000	744.22	0.000	584.02	0.000	411.36

Table 11. Estimates of λ and AIC for the fitted GOLLMax semiparametric regression for scenarios 2a, 2 b and 2c with $n = 200$.

	scenario 2a		scenario 2b		scenario 2c	
$df^{\dagger}$	λ	AIC	λ	AIC	λ	AIC
1	0.183	569.28	0.195	481.12	0.202	358.5
2	0.037	515.49	0.038	434.69	0.038	313.07
3	0.011	492.58	0.012	412.22	0.011	301.67
4	0.004	486.22	0.002	403.32	0.002	297.35
5	**0.002**	**484.91**	**0.001**	**403.26**	**0.002**	**295.54**
6	0.001	484.94	0.001	403.68	0.001	295.94
7	0.001	485.36	0.000	404.25	0.001	296.58
8	0.000	485.90	0.000	404.79	0.000	296.92
9	0.000	486.48	0.000	405.37	0.000	297.03
10	0.000	487.07	0.000	405.52	0.000	297.35

Table 12. Basic statistics of the BMI response.

Mean	Median	SD	Skewness	Kurtosis	Min.	Max.
28.648	28.0	5.967	0.679	0.734	15.0	55.0

values, thus indicating that it is the most suitable regression for simulated data in the presence of non-linear effects.

Tables 10 and 11 report a study where the actual degrees of freedom of adjustment is done by cubic spline for the same previous scenarios. It is noted that when there are behaviors like point cloud or slightly nonlinear as scenarios a and b, the default $df^{\dagger} = 3$ provides good results in smoothing the points. For relationships with sinusoidal forms such as those in scenarios c and d, higher degrees of freedom or the choice of another type of smoothing function may be required.

5. Applications

Two applications show the flexibility of the GOLLMax semiparametric regression to fit uncensored and censored data.

5.1. Application 1: uncensored data

Consider the relationship of two variables in the data analyzed by Kelly et al. (2012), who proposed the development of a tool to predict, in the preoperative period, the need for a patient to attend an extended care service after orthotopic liver transplantation. The response variable is $y =$ BMI and the explanatory variable is $x_1 =$ age of the transplanted patient, where $n = 777$

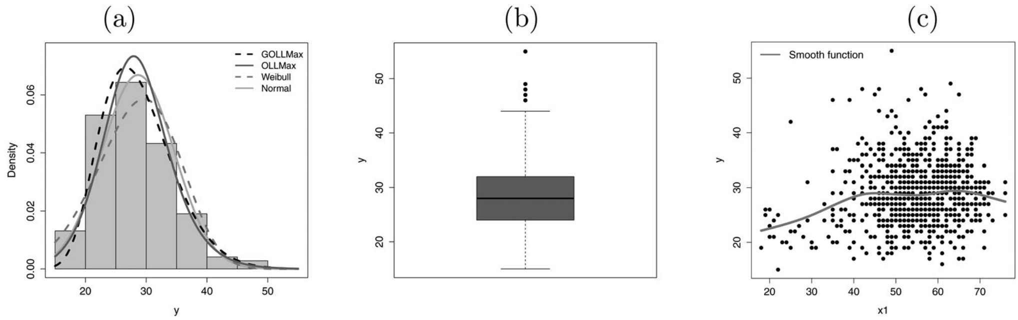

Figure 5. Plots of the BMI data. (a) Histogram and estimated GOLLMax, OLLMax, Weibull and normal densities. (b) Box-plot for the response variable y. (c) Plot of y versus x_1 with the smooth fitted curve.

Table 13. Three measures, df, $df^{\dagger}$ and $\hat{\lambda}$ for the fitted GOLLMax, OLLMax, Weibull and normal semiparametric regressions to the BMI data.

Model	GD	AIC	BIC	df	$df^{\dagger}$	$\hat{\lambda}$
GOLLMax	4904.09	4922.09	4963.99	9	5	0.0056
OLLMax	4905.93	4921.93	4959.17	8	5	0.0056
Weibull	5029.35	5045.35	5082.58	8	5	0.0060
Normal	4949.49	4963.48	4996.07	7	4	0.0121

observations. Table 12 provides basic statistics for the response. The value of the skewness suggests positively skewed distributions to explain these data.

In the following analysis, the GOLLMax, its special OLLMax case, and the Weibull and normal distributions are compared to explain the BMI. The density functions fitted to BMI are reported in Figure 5a. As previously mentioned, the normal distribution is not suitable due to positive asymmetry. Figure 5b shows a box plot for BMI which indicates the presence of asymmetry and possible outliers. The dispersion plot of the observed y against x_1 with the smooth fitted curve and correlation of the variables are given in Figure 5c.

Further, the systematic components (see Sec. 2) from the GOLLMax, OLLMax, Weibull and normal semiparametric regressions are defined as:

$$\begin{aligned}
&\textbf{GOLLMax}: \ \mu_{\mathrm{i}} = \exp\{\beta_{10} + \mathrm{cs}(\mathrm{x}_{\mathrm{i}1}, \mathrm{df}^{\dagger})\}, \quad \sigma = \exp(\beta_{20}) \quad \text{and} \quad \nu = \exp(\beta_{30});\\
&\textbf{OLLMax}: \ \mu_{\mathrm{i}} = \exp\{\beta_{10} + \mathrm{cs}(\mathrm{x}_{\mathrm{i}1}, \mathrm{df}^{\dagger})\} \quad \text{and} \quad \sigma = \exp(\beta_{20});\\
&\textbf{Weibull}: \ \mu_{\mathrm{i}} = \exp\{\beta_{10} + \mathrm{cs}(\mathrm{x}_{\mathrm{i}1}, \mathrm{df}^{\dagger})\} \quad \text{and} \quad \sigma = \exp(\beta_{20});\\
&\textbf{Normal}: \ \mu_{\mathrm{i}} = \beta_{10} + \mathrm{cs}(\mathrm{x}_{\mathrm{i}1}, \mathrm{df}^{\dagger}) \quad \text{and} \quad \sigma = \exp(\beta_{20}).
\end{aligned}$$

The GD, AIC and BIC values, the effective degrees of freedom and estimate of the smoothing parameter λ from the fitted semiparametric regressions to the current data are reported in Table 13. The values of the measures indicate that the GOLLMax and OLLMax semiparametric regressions outperform the Weibull and normal regressions. The likelihood ratio (LR) statistic for comparing the GOLLMax and OLLMax regressions, i.e., testing $H_0 : \nu = 1,$ is $w = 1.835$ (p-value $= 0.175$), which reveals that the null regression can be accepted for these data.

It is shown in Figure 6 how the $df^{\dagger}$ values are chosen. In this way, the parameter λ is estimated according to the choice of $df^{\dagger}$ which minimizes the AIC measure. Table 14 provides the MLEs, SEs and p-values from the fitted OLLMax semiparametric regression to the these data. The intercept corresponding to the parameter of the nonlinear effects is significant at the 5% significance level. However, such value is not interpretable.

The case-deletion measure LD_i is calculated as in Sec. 3.2. The influence measure index plot is reported in Figure 7a. Figure 7b shows the scatter plot with possible influential points. These plots reveal that the observations ♯224, ♯618 and ♯685 are possible influential points. Table 15 provides the relative change of each estimate, say $RC_{\theta_j} = [(\hat{\omega}_j - \hat{\omega}_j(I))/\hat{\omega}_j] \times 100,$ and the associated

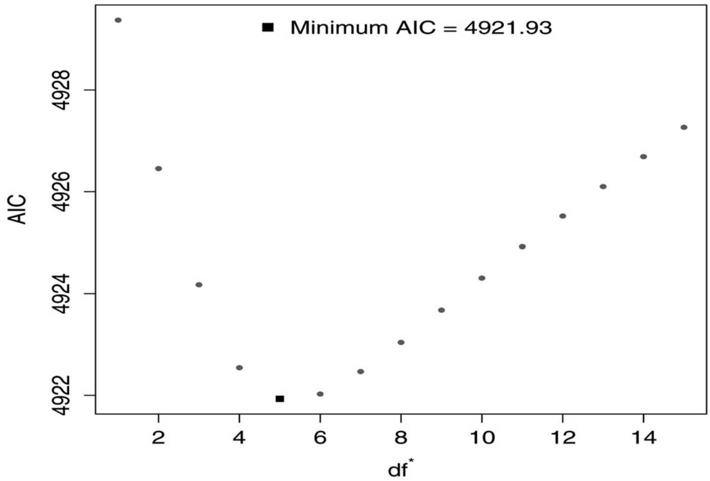

Figure 6. Plots of the AIC versus $df^{\dagger}$ for the OLLMax regression.

Table 14. Estimates and SEs from the fitted OLLMax semiparametric regression.

Parameter	MLE	SE	p-value
β_{10}	3.096	0.038	< 0.001
$cs(x_1, df^{\dagger} = 5.0)$	0.003	0.007	< 0.001
β_{20}	0.876	0.030	–

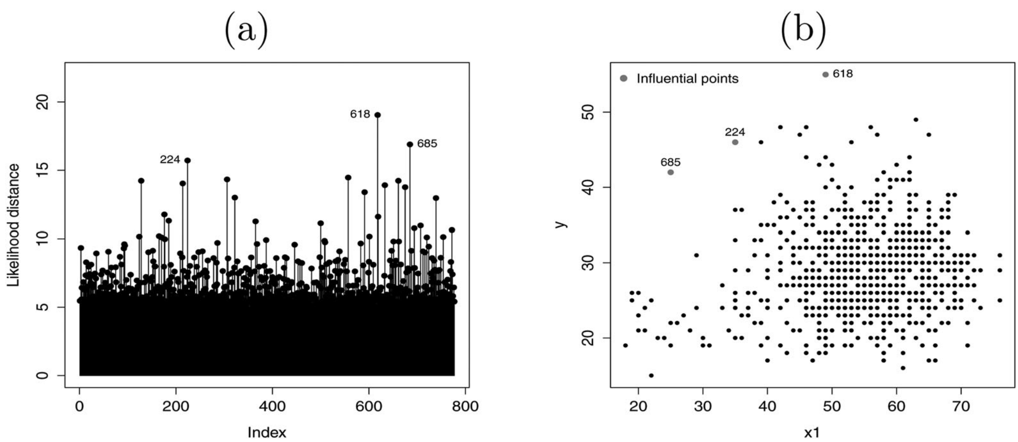

Figure 7. Plots from the fitted OLLMax semiparametric regression to the BMI data. (a) Index plot versus likelihood distance. (b) Observed y versus x_1 with influential points.

Table 15. Estimates, p-values, and their relative changes [-*RC*- in %] for the corresponding set.

Dropping	None	Set I_1	Set I_2	Set I_3
	3.096	3.090	3.102	3.146
β_{10}	(< 0.001)	(<0.001)	(<0.001)	(<0.001)
	–	[0.161]	[-0.193]	[-1.614]
	0.003	0.003	0.003	0.002
$cs(x_1, df^{\dagger} = 5.0)$	(<0.001)	(<0.001)	(<0.001)	(0.003)
	–	[0]	[0]	[33.33]

p-value (in parenthesis), where $\hat{\omega}_j(I)$ is the estimate of ω_j if the set "I" of cases is eliminated. Table 15 gives results for the following sets: $I_1 = \{\sharp 224\}, I_2 = \{\sharp 618\}$ and $I_3 = \{\sharp 685\}$.

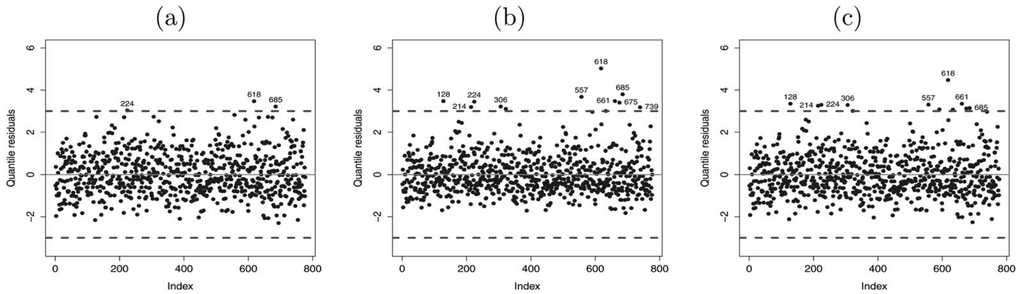

Figure 8. The index plot of the quantile residuals with range $[-3, 3]$ from three fitted semiparametric regressions. (a) OLLMax. (b) Weibull. (c) Normal.

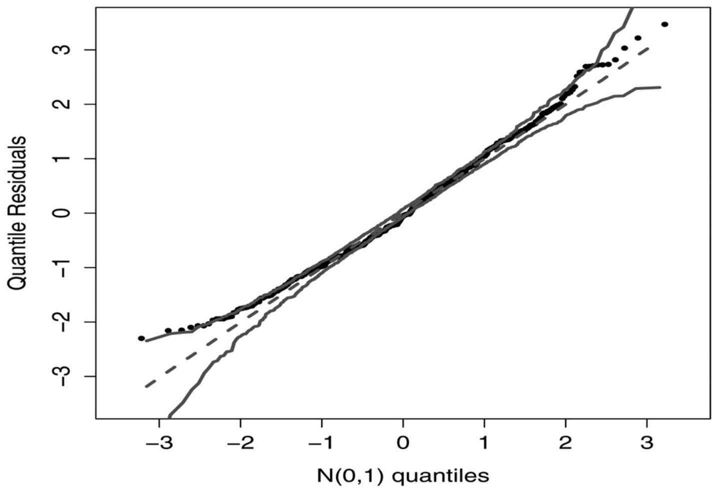

Figure 9. Normal probability plot for the quantile residuals with envelope from the fitted OLLMax semiparametric regression model to BMI data.

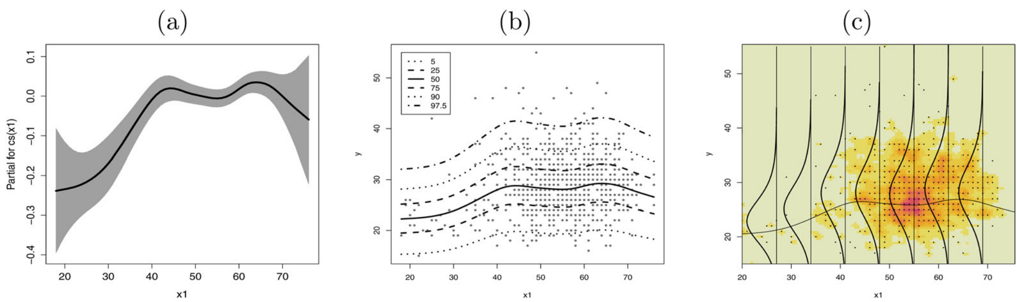

Figure 10. The OLLMax semiparametric regression model fitted for the BMI data. (a) Fitted partial effects of x_1. (b) Fitted percentile curves for $qx100 = (5, 25, 50, 75, 90, 97.5)$ versus x_1. (c) Smoothed scatter plot showing the fitted conditional distribution of BMI changing with x_1.

The numbers in Table 15 indicate that these observations have little influence on the estimates of the parameters of the OLLMax semiparametric regression. The plots of the residuals rq_i (see Sec. 3.2) versus the index of the observations from the fitted OLLMax, Weibull and normal semiparametric regressions are displayed in Figure 8a-c, respectively. In Figure 8a, all points are included in the interval $[-3, 3]$ except the cases ♯224, ♯618 and ♯685, and show that the residuals

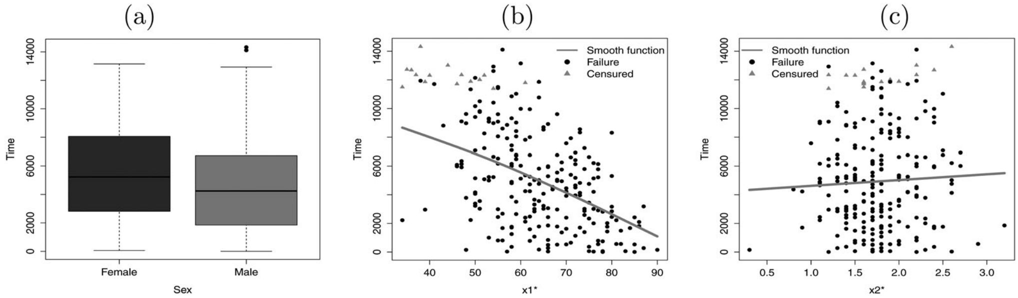

Figure 11. Plots for MGUS data. (a) Box plot by x_3^*. (b) Observed y against x_1^* with fitted smooth curve for uncensored values. (c) Observed y against x_2^* with fitted smooth curve for uncensored values.

behave randomly. Then, there is no evidence against the regression assumptions. The plots in Figures 8b and c reveal many discrepant points.

The plots of the qrs and the envelope from the fitted OLLMax semiparametric regression are displayed in Figure 9, where there are no discrepant observations. So, the fitted regression is adequate for these data.

Figure 10a indicates that the BMI increases rapidly until the 40 year of age, and then remains approximately constant from 40 to 60 years of age, and declines for age > 60. Figure 10b gives six fitted percentile curves q x $100 = (5, 25, 50, 75, 90, 97.5)$ for y versus x_1. Figure 10c exhibits the functional curve fitted by the OLLMax semiparametric regression to the BMI data with some marginal densities for different values of x_1. In summary, this regression can be chosen as the best one to explain the current data.

5.2. Application 2: censored data

In *monoclonal gammopathy of undetermined significance* (MGUS), blood samples contain abnormal proteins called monoclonal (M) proteins. Plasma cells in the bone marrow (the soft substance at the center of bones) produce M proteins. Here, the 241 MGUS observations included in the survival library in R software are used; for details of these data, see Kyle (1993). The response and explanatory variables are (for $i = 1, \ldots, 240$):

- y_i: days from diagnosis to the last follow-up;
- x_{i1}^* : age in years at the detection of MGUS;
- x_{i2}^* : size of the monoclonal protein spike at diagnosis;
- x_{i3}^* : sex, a factor with level male and reference female,

since the variable x_{i2}^* contains a missing observation. Note that x_1^* or x_2^* can have nonlinear effects on the response variable, so we will examine later on the effects of these variables on the response variable.

The box plot for x_3^* given in Figure 11a reveals that there is some similarity of the failure times between males and females. Figure 11b and c display the dispersion plot of y versus the covariates x_1^* and x_2^*, respectively.

First, consider an analysis on the survival and censoring times without covariates. The TTT plot in Figure 12a indicates that the hrf of the data has a decreasing shape. So, the GOLLMax, OLLMax and Weibull distributions can be chosen to model these data. The estimated survival functions of the GOLLMax, OLLMax and Weibull distributions and the empirical survival function are displayed in Figure 12b, where the estimated hrfs are given in Figure 12c. Hence, the GOLLMax distribution provides a good fit to the time from diagnosis to the last follow-up.

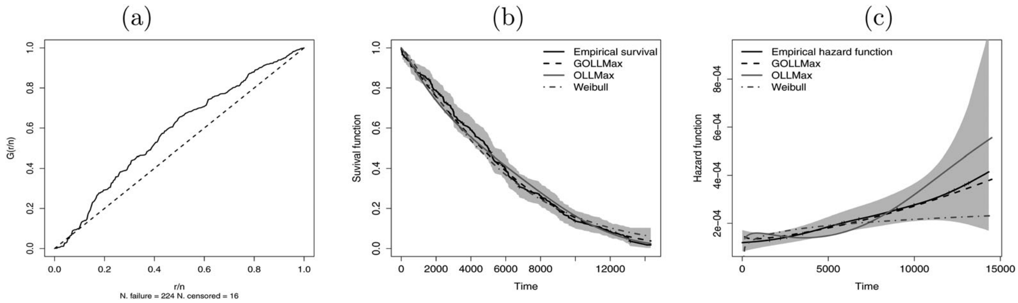

Figure 12. Plots for MGUS data. (a) TTT-plot for y. (b) Empirical and estimated survival function by the GOLLMax, OLLMax and Weibull models. (c) Empirical and estimated hazard function by the GOLLMax, OLLMax and Weibull models.

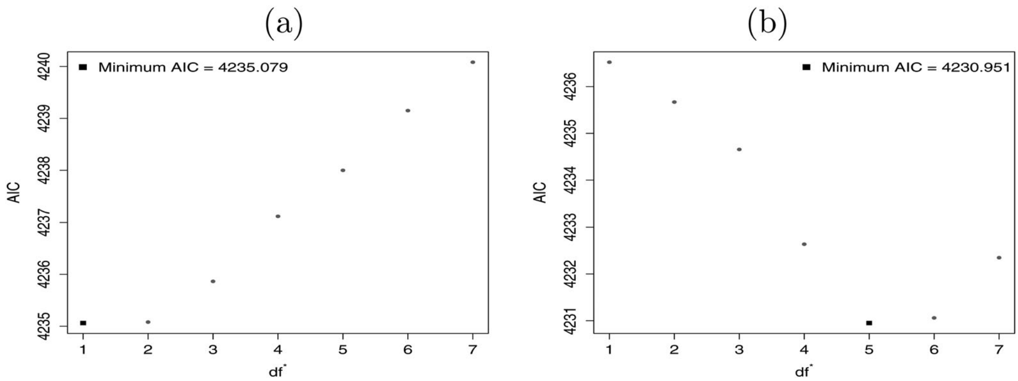

Figure 13. Plots of the AIC versus $df^{\dagger}$ for the GOLLMax regression.

Second, consider two semiparametric regressions with three continuous covariates. The values of $df^{\dagger}$ and λ are calculated as follows:

i. For obtaining $df^{\dagger}$ and $\hat{\lambda}$ with respect to x_1, consider x_2^* and x_3^* fixed with the systematic component: $\mu_i = \exp\{\beta_{10} + cs(x_{i1}, df^{\dagger}) + \beta_{12}x_{i2}^* + \beta_{13}x_{i3}^*\}$;
ii. For obtaining $df^{\dagger}$ and $\hat{\lambda}$ with respect to x_2, consider x_1^* and x_3^* fixed with the systematic component: $\mu_i = \exp\{\beta_{10} + \beta_{11}x_{i1}^* + cs(x_{i2}, df^{\dagger}) + \beta_{13}x_{i3}^*\}$.

Figure 13 reveals how the $df^{\dagger}$ values are chosen for the GOLLMax semiparametric regression. According to the values reported in Figure 13a from the systematic component (i), the model for variable x_1^* can be considered linear in agreement with the initial exploratory shown in Figure 11b. From the systematic component (ii) in Figure 13b there is a minimum point thus indicating the need to use a smoothing function for variable x_2. Thus, to maintain our usual notation from now on, x_2^* will be considered as x_2. The same procedure applies for the OLLMax and Weibull regressions, respectively.

Further, the GOLLMax, OLLMax and Weibull semiparametric regressions are adopted for these data with the systematic components:

$$\textbf{GOLLMax}\begin{cases} \mu_{\rm i} = \exp\{\beta_{10} + \beta_{11}{\rm x}_{\rm i1}^* + {\rm cs}({\rm x}_{\rm i2}, {\rm df}^{\dagger}) + \beta_{13}{\rm x}_{\rm i3}^*\}, \\ \sigma = \exp(\beta_{20}) \quad \text{and} \quad \nu = \exp(\beta_{30}); \end{cases}$$

$$\textbf{OLLMax}: \ \mu_{\rm i} = \exp\{\beta_{10} + \beta_{11}{\rm x}_{\rm i1}^* + {\rm cs}({\rm x}_{\rm i2}, {\rm df}^{\dagger}) + \beta_{13}{\rm x}_{\rm i3}^*\} \quad \text{and} \quad \sigma = \exp(\beta_{20});$$

$$\textbf{Weibull}: \ \mu_{\rm i} = \exp\{\beta_{10} + \beta_{11}{\rm x}_{\rm i1}^* + {\rm cs}({\rm x}_{\rm i2}, {\rm df}^{\dagger}) + \beta_{13}{\rm x}_{\rm i3}^*\} \quad \text{and} \quad \sigma = \exp(\beta_{20}).$$

Table 16. Three measures, *df*, *df*† and $\hat{\lambda}$ for the fitted GOLLMax, OLLMax and Weibull semiparametric regressions to the MGUS data.

Regression	GD	AIC	BIC	*df*	*df*†	$\hat{\lambda}$
GOLLMax	4208.95	4230.95	4269.24	11	5	0.00095
OLLMax	4216.59	4236.59	4271.40	10	5	0.00098
Weibull	4226.56	4248.56	4286.86	11	6	0.00085

Table 17. Estimates and SEs from the fitted GOLLMax semiparametric regression to the MGUS data.

Parameter	MLE	SE	*p*-value
β_{10}	10.899	0.459	<0.001
β_{11}	−0.037	0.005	<0.001
$cs(x_2, df^\dagger = 5.0)$	0.099	0.150	0.509
β_{13}	−0.131	0.117	0.265
β_{20}	−0.657	0.058	–
β_{30}	−0.261	0.050	–

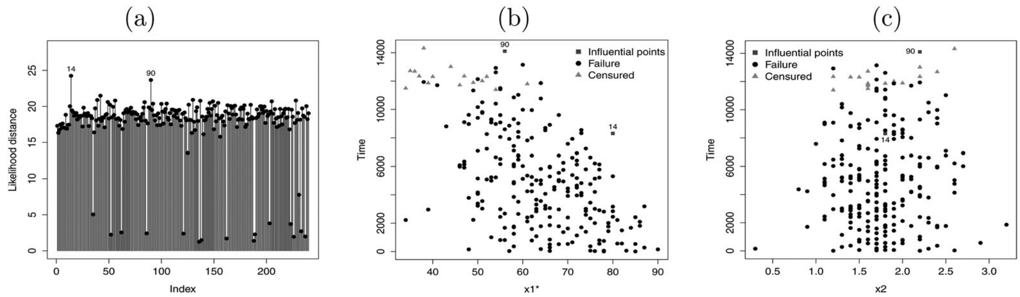

Figure 14. Plots for the fitted OLLMax semiparametric regression model to the BMI data. (a) Index plot versus likelihood distance. (b) Observed *y* versus x_1^* with influential points. (c) Observed *y* versus x_2 with influential points.

The values of some statistics and the effective degrees of freedom and the estimate of the smoothing parameter λ from the fitted semiparametric regressions are reported in Table 16. By comparing the figures in this table, it is clear that the GOLLMax semiparametric regression outperforms the OLLMax and Weibull regressions for all criteria. Hence, the wider regression can be used effectively in the analysis of these censoring data.

For comparing the GOLLMax and OLLMax regressions, i.e., for testing the null hypothesis $H_0 : \nu = 1$, the LR statistic is $w = 7.645$ (p-value $= 0.005$) thus indicating that the GOLLMax semiparametric regression could be chosen for these data.

Table 17 reports the MLEs, SEs and p-values from the fitted GOLLMax semiparametric regression to the MGUS data. The numbers in this table reveal (at the 5% significance level) that the intercept related to the nonlinear effects is significant. The coefficients of the linear term for variable x_1^* is also significant, i.e., the age has an effect on the lifetime of patients. Due to the linear decreasing trend, the time from diagnosis to the last follow-up decreases when the age of the patient increases. The coefficients of the nonlinear term is presented. However, such values are not interpretable. The variable x_3^* (sex) also has no significant effect in relation to the lifetime of the patients under study.

The case-deletion measure LD_i is calculated and the index plots are displayed in Figure 14a. Figure 14b and Figure 14(c) reveal the scatter plot with possible influential points identified for the variables x_1^* and x_2, respectively.

These plots reveal that the observations $\sharp$14 and$\sharp$90 are possible influential points. Table 18 gives the relative change (in percentage) of each estimate defined by $RC_{\boldsymbol{\theta}_j} = [(\hat{\boldsymbol{\omega}}_j - \hat{\boldsymbol{\omega}}_j(I))/\hat{\boldsymbol{\omega}}_j] \times 100$, and the corresponding p-value, where $\hat{\boldsymbol{\omega}}_j(I)$ is the MLE of $\boldsymbol{\omega}_j$ after the "set I" of observations being removed. Table 15 provides the following sets: $I_1 = \{\sharp 14\}$ and $I_2 = \{\sharp 90\}$.

Table 18. MLEs, p-values (in parentheses), and their relative changes [-*RC*- in %] for the corresponding set.

Dropping	None	Set I_1	Set I_2
	10.899	11.011	10.798
β_{10}	(< 0.001)	(<0.001)	(<0.001)
	–	[−1.027]	[0.917]
	−0.037	−0.039	−0.037
β_{11}	(< 0.001)	(<0.001)	(<0.001)
	–	[−8.108]	[0]
	0.099	0.094	0.106
$cs(x_2, df^{\dagger} = 5.0)$	(0.509)	(0.520)	(0.497)
	–	[5.050]	[−7.070]
	−0.131	−0.156	−0.145
β_{13}	(0.265)	(0.173)	(0.222)
	–	[−19.847]	[−11.450]

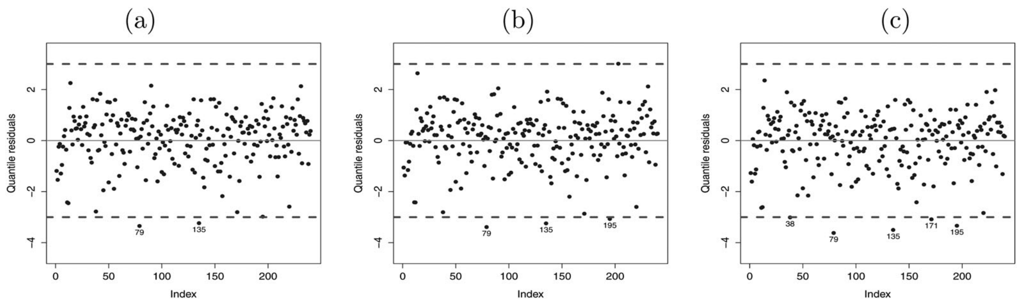

Figure 15. Plots of the qrs versus the index of the observations. (a) GOLLMax regression. (b) OLLMax regression. (c) Weibull regression.

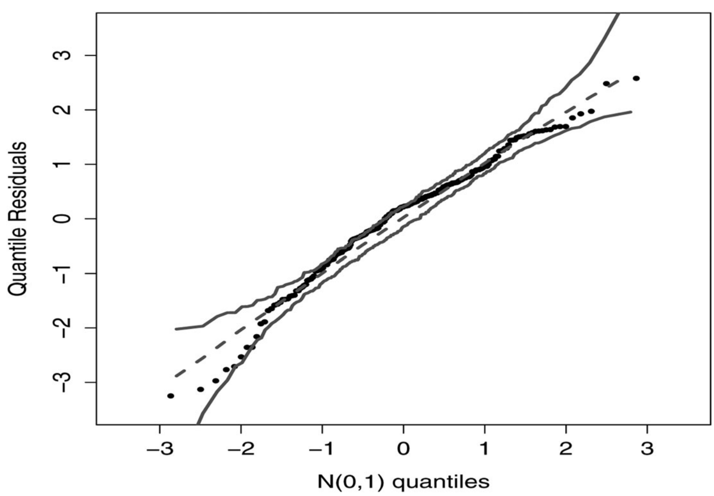

Figure 16. Normal probability plot for the qrs with envelope from the fitted GOLLMax semiparametric regression to MGUS data.

Based on the figures in Table 18, the estimates for the GOLLMax semiparametric regression are robust for the deletion of influential observations. Moreover, the significance of these estimates does not change (at the 5% significance level) after eliminating these observations.

The plots of the residuals versus the observation index for three fitted regressions are reported in Figures 15a, b and Figure 15(c), respectively. The plot in Figure 15a indicates that the residuals appear to behave randomly and that all of them are included in the interval $[-3, 3]$ except the cases ♯79 and ♯135. Then, there is no evidence that the model assumptions are inadequate. However, there are more discrepant points in Figures 15b and c.

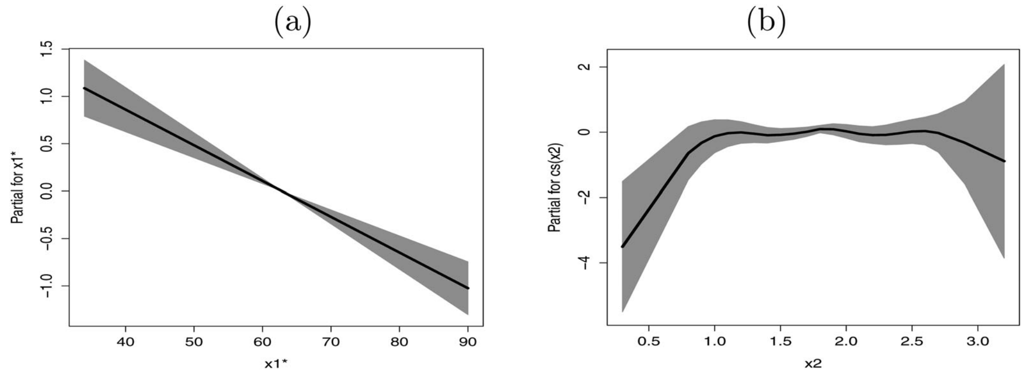

Figure 17. The OLLMax semiparametric regression fitted to the MGUS data. (a) Fitted partial effects of x_1^*. (b) Fitted partial effects of x_2.

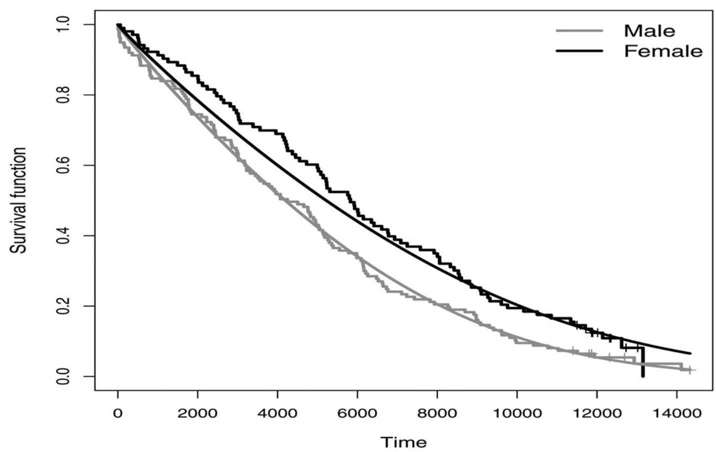

Figure 18. Estimated and empirical survival functions for sex covariate.

The plots of the residuals and the simulated envelope (Atkinson 1985) for the GOLLMax semiparametric regression in Figure 16 reveal the absence of discrepant observations and that this regression is adequate to explain the current data.

The partial effects for the covariates in relation to the systematic components are displayed in Figure 17. The effect of the linear term for variable x_1^* is presented in Figure 17a. As interpreted before, the lifetime decreases when the age of the patient increases. Figure 17b reveals that for M protein spike measurements between approximately 0.5 and 1 there is a growing linear trend in lifetime. For values between 1 and 2.6, the lifetime remains constant and from 2.6 there is a linear decreasing trend in the lifetime.

Finally, in order to assess if the model is appropriate, the empirical and estimated survival functions of the ESC regression are plotted in Figure 18 for the sex covariate. We may conclude from the plots that the OLLMax semiparametric regression provides a suitable fit to the current data.

6. Concluding remarks

We proposed the *generalized odd log-logistic Maxwell* semiparametric regression which is very flexible for modeling response variable with positive support. Procedures for fitting this regression and for model diagnostics are included in the gamlss package reported in the Appendix. Two real data sets are used to illustrate the importance of the proposed regression for censored and uncensored data.

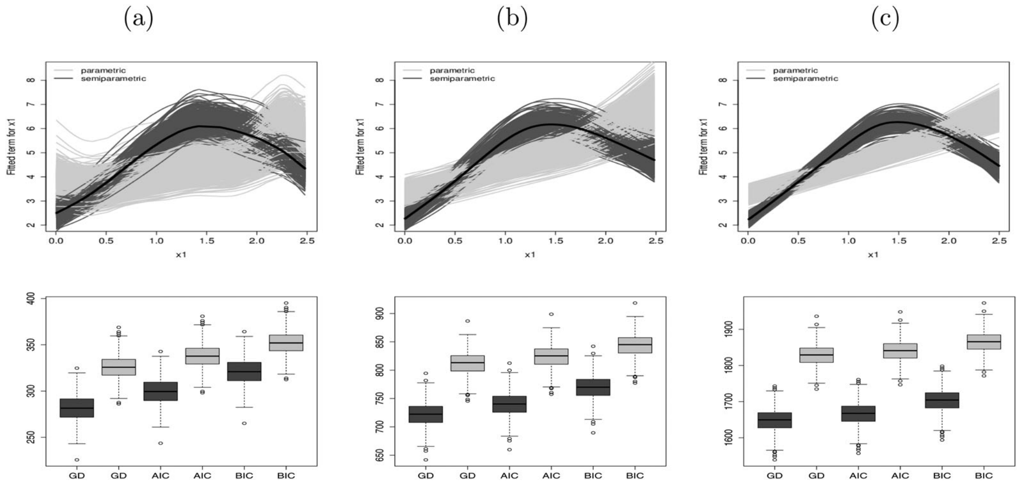

Figure 19. The fitted GOLLMaxSemi regression under scenario 1a for the effect of x_1 to the parameter μ. (a) $n = 80$. (b) $n = 200$. (c) $n = 450$.

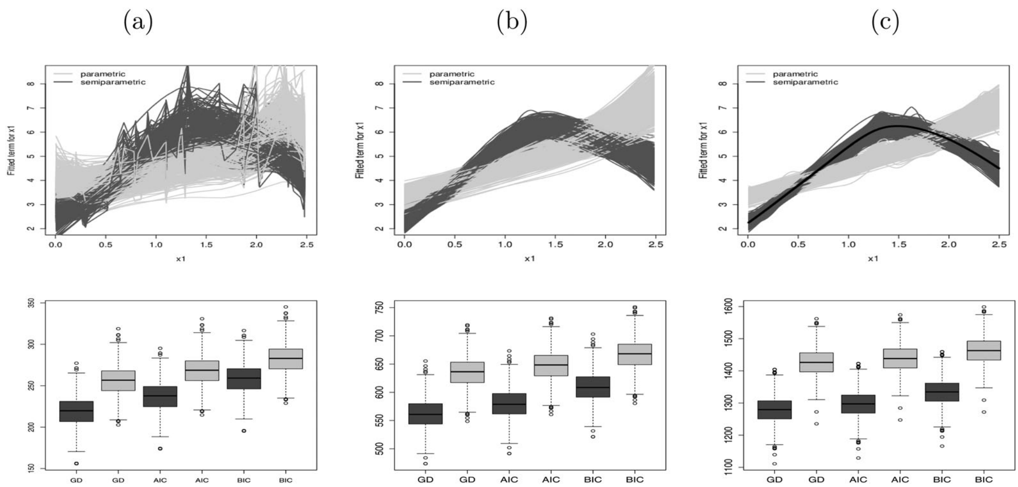

Figure 20. The fitted GOLLMaxSemi regression under scenario 1b for the effect of x_1 to the parameter μ. (a) $n = 80$. (b) $n = 200$. (c) $n = 450$.

Appendix

Here, we present the codes implemented in the GAMLSS package and optim in the R software, respectively. The pdf, cdf, qf and the samples generator functions are:

```
#required packages
library(flexsurv); library(numDeriv); library(gamlss);
library(gamlss.cens);

source('https://gist.githubusercontent.com/fabiopviera/
bf29416bae31134130764a096b41e7ce/raw/1e76c9a92c7c17f8b0bdff98ca53c900ce8-
d3ea6/GOLLMax.r')

#implemented codes dGOLLMax(x,mu,sigma,nu,tau) #pdf
pGOLLMax(x,mu,sigma,nu,tau) #cdf
qGOLLMax(u,mu,sigma,nu,tau) #qf
rGOLLMax(n,mu,sigma,nu,tau) #samples generator
```

The code used to obtain the estimates of the semiparametric regression in Application 1:

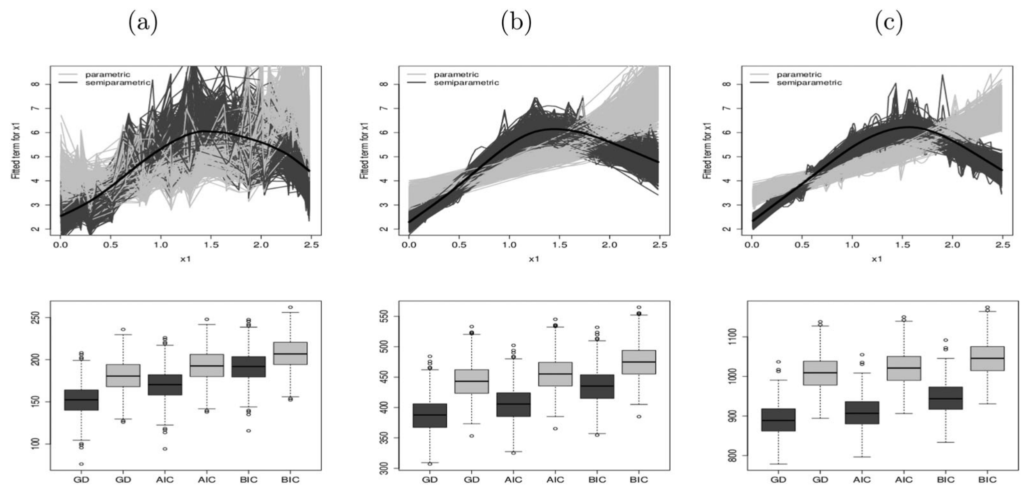

Figure 21. The fitted GOLLMaxSemi regression under scenario 1c for the effect of x_1 to the parameter μ. (a) $n = 80$. (b) $n = 200$. (c) $n = 450$.

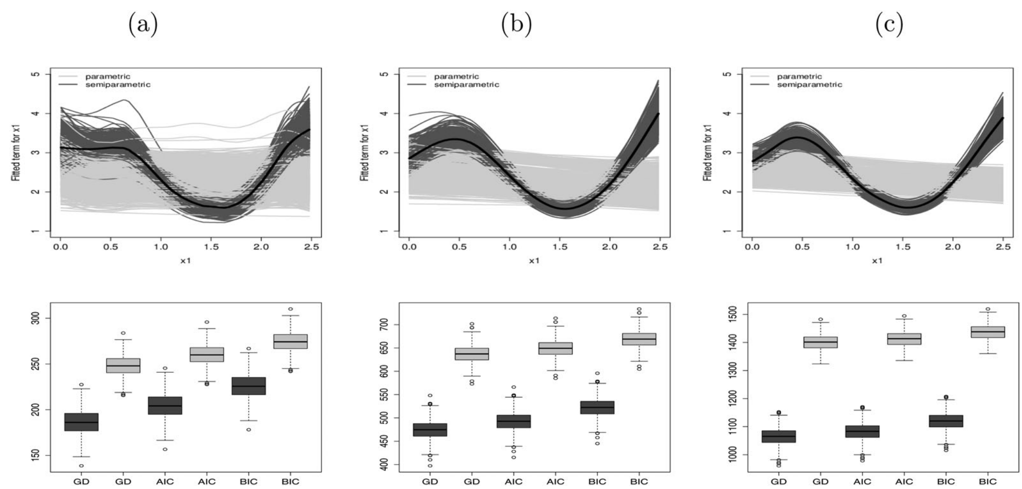

Figure 22. The fitted GOLLMaxSemi regression under scenario 2a for the effect of x_1 to the parameter μ. (a) $n = 80$. (b) $n = 200$. (c) $n = 450$.

```
fit1=gamlss(y~cs(x1,df = 5),sigma.formula =~1,nu.formula =~1,
family=GOLLMax(),n.cyc = 200,c.crit = 0.01,
data=data1)
```

The code used to obtain the estimates of the semiparametric regression in Application 2:

```
fit2=gamlss(Surv(y,delta)~cs(x1,df = 5)+x2+x5,sigma.formula
=~1,nu.formula =~1, family=cens(GOLLMax()),n.cyc = 200,c.crit = 0.01,
data=data2)
```

The code used by optim for the parametric regression modeling:

```
verosi <- function(vpar){
p1 <- vpar[1]
p2 <- vpar[2]
p3 <- vpar[3]
p4 <- vpar[4]
p5 <- vpar[5]

mu <- exp(p1+p2*x1+ p3*x2)
sigma <- exp(p4)
nu <- exp(p5)
```

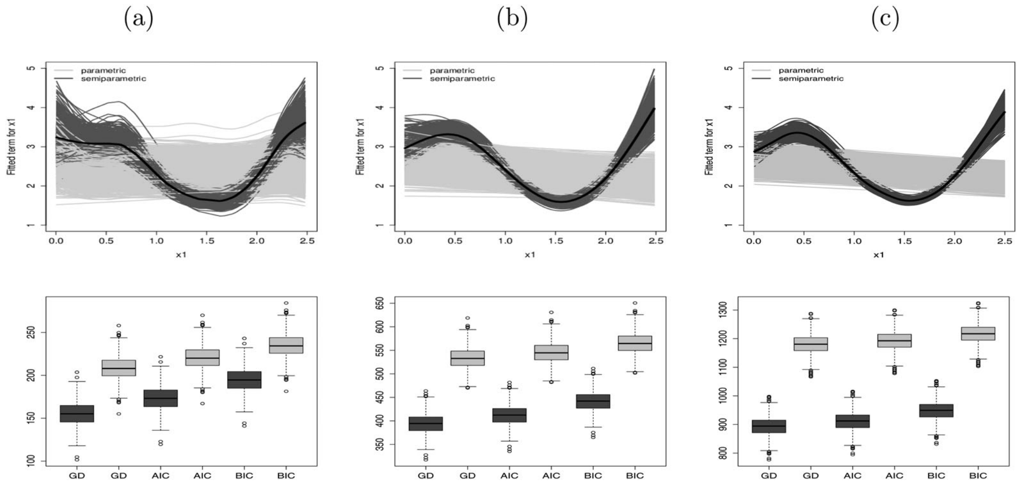

Figure 23. The fitted GOLLMaxSemi regression under scenario 2b for the effect of x_1 to the parameter μ. (a) $n=80$. (b) $n=200$. (c) $n=450$.

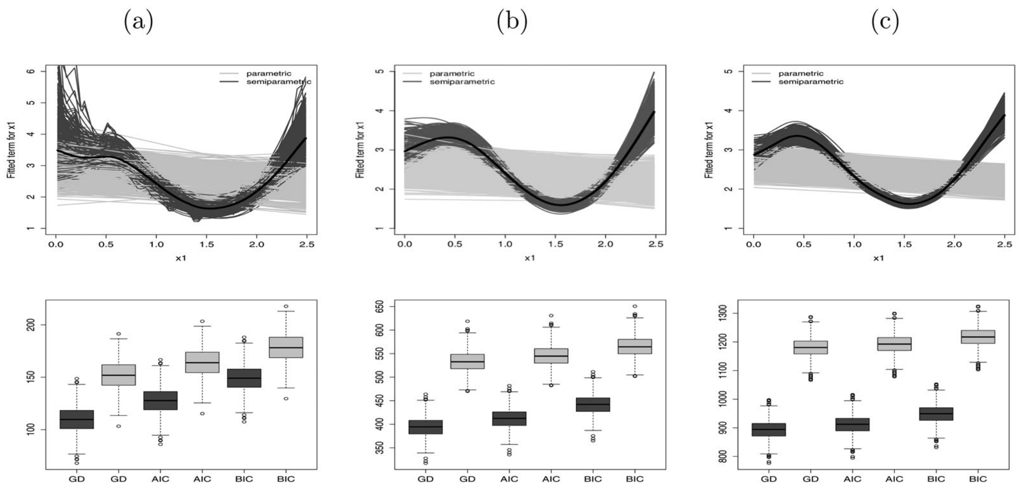

Figure 24. The fitted GOLLMaxSemi regression under scenario 2c for the effect of x_1 to the parameter μ. (a) $n=80$. (b) $n=200$. (c) $n=450$.

```
g <- dgengamma.orig(t,shape = 2,scale=mu,k=3/2)
G <- pgengamma.orig(t,shape = 2,scale=mu,k=3/2)
FF <- (G^(sigma*nu))/(((G^(sigma*nu))+(1-G^sigma)^nu))
S <- 1-FF
f <- (sigma*nu*g*(G^(sigma*nu-1)) *(1-G^sigma)^(nu-1))/(((G^(sigma*nu)) +
(1-G^sigma)^(nu))^2)

lv <- delta*log(f)+(1-delta)*log(S)
logv <- sum(lv)
return(logv)
}

valorin <- c(a10,a11,a12,b20,c30)

result=optim(valorin,verosi, hessian=TRUE, control=list(fnscale = -1),
method = "BFGS");result
```

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